

IE410: Introduction to Robotics – Winter 2026

Project – Part B: Simulation of a Theo Jansen Mechanism

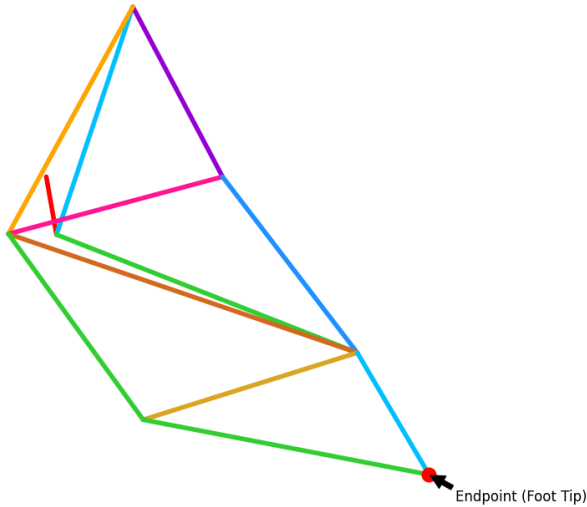
1 Overview and Objectives

The objective of this project is to model and simulate a single degree-of-freedom (DoF) Theo Jansen walking mechanism. Driven by a single rotary input, this 12-link planar mechanism generates a natural, human-like ankle trajectory. This report details the kinematic construction, the extracted endpoint trajectory, and a sensitivity analysis of specific link dimensions to demonstrate its viability for cost-effective robotic gait rehabilitation.

2 Modeled Linkage and Gait Trajectory

The mechanism utilizes a standard Theo Jansen planar linkage driven by a constant angular velocity at the input crank. The designated end-effector (the “foot”) is located at the lowest distal joint of the lower triangular assembly, intended to trace the path of a human ankle in the sagittal plane.

1. Modeled Linkage & Endpoint



2. Generated Gait Trajectory

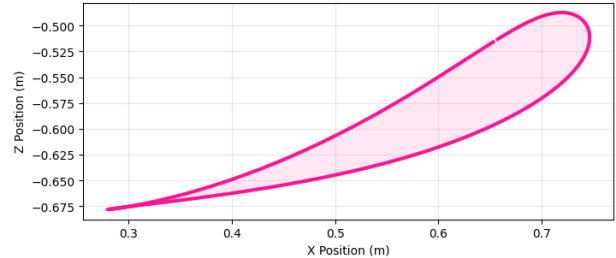


Figure 2: Endpoint trajectory over one cycle.

Figure 1: Schematic of the 12-link mechanism.

As seen in Figure 2, the extracted Cartesian trajectory clearly resembles a natural gait curve. It features a flatter bottom portion representing the **stance phase** (steady ground contact for forward propulsion) and an arched upper portion representing the **swing phase** (foot clearance).

3 Effect of Link Dimensions

To understand how linkage geometry dictates the output path to fit varying human anthropometrics, parametric sweeps were conducted on the lower rocker and the crank.

- **Lower Rocker (L8):** Altering L8 from 35.5 cm to 47.5 cm (Figure 4) drastically changes the spatial footprint. Lengthening this link stretches the curve, influencing the step height and the angle of attack during the pre-swing phase.

Mechanism at Six Crank Positions

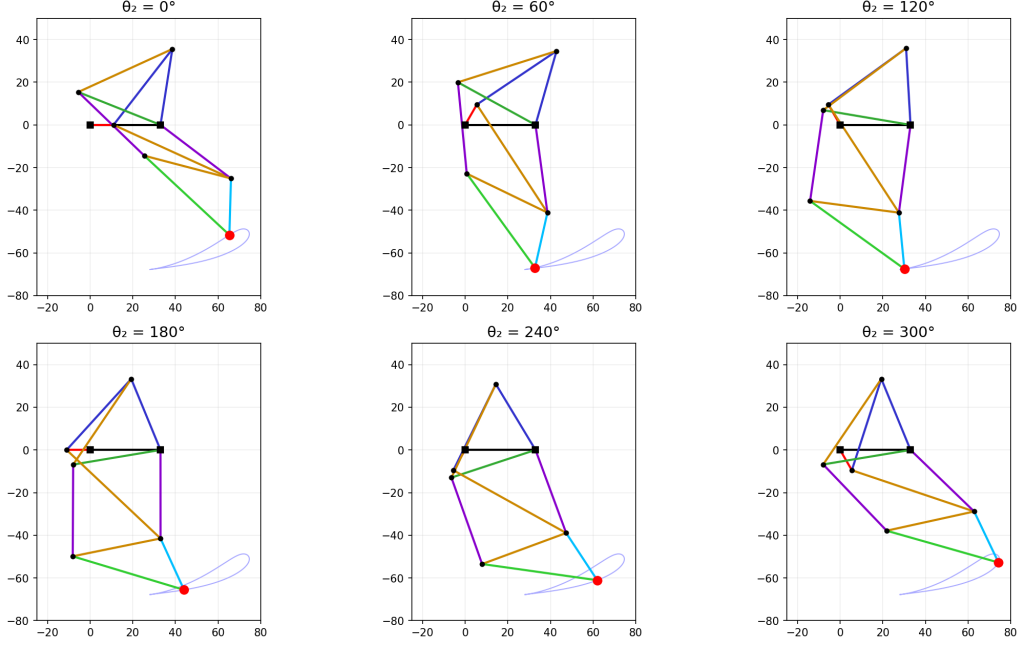


Figure 3: Snapshots of the mechanism's fluid motion across six crank angles (0° to 300°).

3. Effect of Link Dimensions (L8 Variation)

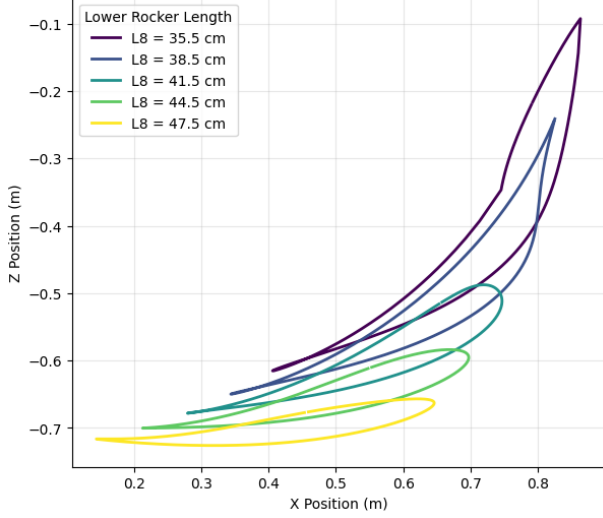


Figure 4: Effect of Lower Rocker (L8) variation.

4. Effect of Crank Dimension (L1 Variation)

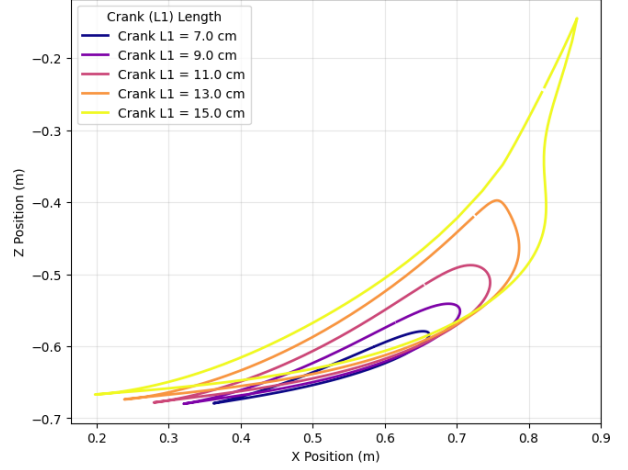


Figure 5: Effect of Crank Length (L1) variation.

- **Crank Length (L1):** Varying the crank radius from 7.0 cm to 15.0 cm (Figure 5) scales the overall amplitude. A larger crank radius expands the trajectory globally, increasing both horizontal stride length and vertical step height proportionally.

4 Comparison with Natural Walking

The simulated curves successfully approximate the natural human walking "meta-trajectories" found in biomechanical literature. The characteristic teardrop shape mirrors the path of a healthy human ankle. The flat bottom ensures natural ground contact, which is vital for providing proprioceptive feedback, while the single rotary input completely eliminates the need for complex, variable-speed multi-motor control systems.